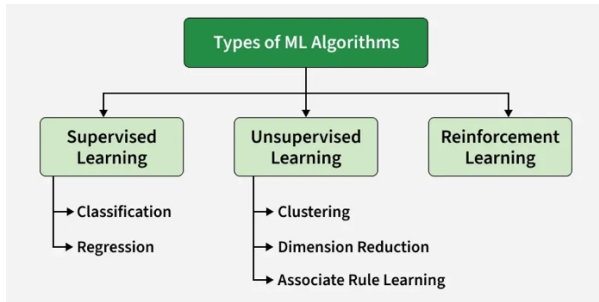


Types of Machine Learning



Supervised Learning

Supervised learning is a type of Machine Learning where the model learns from **labeled data**.

- **Input (X):** Features or independent variables
- **Output (Y):** Correct answer or target value

The algorithm learns the relationship between **X** and **Y** so that it can predict the output for **new, unseen data**.

What is Classification?

Classification is a supervised learning technique used when the **output (Y) is a category (class or label)** rather than a numerical value.

Definition

Classification is a supervised machine learning technique that learns a function $f(x)$ from labeled data to predict the correct categorical class for previously unseen inputs as accurately as possible.

The goal is:

"Previously unseen records should be assigned a class as accurately as possible."

To assign the correct class (category) to previously unseen records as accurately as possible.

Examples of class labels:

- Spam / Not Spam
- Pass / Fail

- Disease / No Disease
- Fraud / Genuine Transaction
- Cat / Dog
- Yes / No

e.g.

Suppose we have the following training data:

Email Contains Offer?	Unknown Sender?	Class
Yes	Yes	Spam
No	No	Not Spam
Yes	No	Spam
No	Yes	Not Spam

Now a **new email** arrives:

- Contains Offer = Yes
- Unknown Sender = Yes

The model has **never seen this exact email before**, but based on the learned patterns it predicts:

Class = Spam

This is called **classification**.

Mathematical Representation

Given training examples

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

where

- x_i = input/features
- y_i = class label

Example

x (Hours Studied)	y (Result)
2	Fail
4	Pass
5	Pass
1	Fail

The machine learns a function

$$f(x)$$

such that

$$y = f(x)$$

where the output is a **class label**.

Meaning of "Learn a function $f(x)$ "

The algorithm learns a rule that maps input values to categories.

Example

If

- Hours Studied = 6

then

$$f(6) = \text{Pass}$$

If

- Hours Studied = 1

then

$$f(1) = \text{Fail}$$

Here the output is **Pass** or **Fail**, not a number.

Real Life Examples of Classification

Input (X)	Predicted Class (Y)
Email content	Spam / Not Spam
Medical symptoms	Disease / No Disease
Loan application	Approved / Rejected
Bank transaction	Fraud / Genuine
Student data	Pass / Fail
Image	Cat / Dog

Types of Classification

1. Binary Classification

There are **only two classes**.

Examples:

- Yes / No
- Pass / Fail
- Spam / Not Spam
- Fraud / Genuine

2. Multi Class Classification

There are **more than two classes**.

Example:

Recognizing handwritten digits

Classes:

0,1,2,3,4,5,6,7,8,9

Another example:

Image	Class
Apple image	Apple
Orange image	Orange
Banana image	Banana

3. Multi Label Classification

A single record can belong to **multiple classes at the same time**.

Example:

Movie genres

Movie:

- Action
- Thriller
- Suspense

The movie belongs to all three categories simultaneously.

Classification

- **Classification is a supervised learning technique.**
- It uses **labeled training data** (input output pairs).
- The goal is to **predict categorical class labels** for new data.
- The algorithm learns a function $f(x)$ that maps inputs to categories.
- **y is categorical**, meaning it belongs to a predefined set of classes (e.g., Yes/No, Spam/Not Spam).
- Classification is widely used in email filtering, medical diagnosis, image recognition, fraud detection, sentiment analysis, and customer segmentation.

What is Regression?

Regression is a supervised learning technique used when the **output (Y) is a continuous numerical value** rather than a category. To assign the most accurate numerical value to previously unseen records.

Definition Regression is a supervised machine learning technique that learns a function $f(x)$ from labeled data to predict continuous real valued outputs for previously unseen inputs as accurately as possible.

"Previously unseen records should be assigned a value as accurately as possible."

e.g.

Suppose we have historical data:

House Size (sq.ft.)	Price (Lakhs)
1000	40
1200	48
1500	60
1800	72

Now someone asks:

"What will be the price of a **1600 sq.ft.** house?"

The model has **never seen** this house before.

It uses the learned relationship to predict:

Predicted Price = ₹64 Lakhs (approximately)

This is called **regression prediction**.

Mathematical Representation

Given training examples

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

where

- x_i = input/features

- y_i = corresponding output

Example

x (Hours Studied)	y (Marks)
2	35
4	50
6	70
8	90

The machine learns a function

$$f(x)$$

such that

$$y = f(x)$$

This function predicts marks from study hours.

Meaning of "Learn a function $f(x)$ "

The machine tries to discover the mathematical relationship between input and output.

Example:

Suppose it learns

$$f(x) = 10x + 15$$

where

- x = study hours
- y = marks

For a student studying **5 hours**

$$f(5) = 10(5) + 15 = 65$$

Predicted marks = **65**

Regression predicts these kinds of outputs.

Unlike classification, regression does **not** predict labels like:

- Pass/Fail
- Yes/No
- Cat/Dog

Instead, it predicts numbers.

Visual Representation

Difference Between Regression and Classification

Regression	Classification
Predicts numbers	Predicts categories
Output is continuous	Output is discrete
Example: ₹45 Lakhs	Example: Spam/Not Spam
Example: marks = 65	Example: Pass/Fail

Regression

- **Regression is a supervised learning technique.**
- It uses **labeled training data** (input output pairs).
- The objective is to **predict continuous numerical values**.
- The algorithm learns a function $f(x)$ that maps inputs to outputs.
- **y is real valued**, meaning it can take any numerical value (including decimals).
- Regression is widely used in forecasting, finance, healthcare, weather prediction, sales prediction, and price estimation.

Algorithms That Do Both (Classification & Regression)

- Decision Trees:
- Random Forest
- Support Vector Machines (SVM)
- K Nearest Neighbors (KNN)
- Gradient Boosting Machines (GBM / XGBoost / LightGBM)
- Artificial Neural Networks (Deep Learning)

Pure Classification Algorithms

- Logistic Regression
- Naïve Bayes
- Linear Discriminant Analysis (LDA)

Pure Regression Algorithms

- Linear Regression
- Polynomial Regression
- Ridge & Lasso Regression

Unsupervised Learning

What is Unsupervised Learning?

Unsupervised learning is a type of Machine Learning in which the algorithm is trained using **unlabeled data**. Unlike supervised learning, there are **no predefined output labels (Y)**. The algorithm tries to discover **hidden patterns, relationships, or structures** within the data.

Supervised vs. Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labeled data	Uses unlabeled data
Input (X) and Output (Y) are given	Only Input (X) is given
Predicts output values or classes	Finds hidden patterns or groups
Example: Predict house price	Example: Group customers with similar buying habits

1. Clustering

Definition

Clustering is an unsupervised learning technique that groups **similar data points into clusters** based on their characteristics. Data points in the same cluster are more similar to each other than to those in other clusters.

There are **no predefined labels**; the algorithm automatically discovers the groups.

Goal

To divide data into meaningful groups based on similarity.

Example 1: Student Grouping

Suppose a school has information about students:

Student	Math Marks	Science Marks
A	95	92
B	90	88
C	40	45
D	42	38
E	75	78

The clustering algorithm may form:

- **Cluster 1:** A, B (High performers)
- **Cluster 2:** C, D (Low performers)
- **Cluster 3:** E (Average performer)

Notice that nobody told the algorithm these groups it discovered them automatically.

Applications of Clustering

- Customer segmentation
- Image segmentation
- Document grouping
- Social network analysis
- Disease classification
- Market research

- Fraud detection (grouping unusual behavior)

Popular Clustering Algorithms

- K Means Clustering
- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Models (GMM)

2. Dimensionality Reduction

Definition

Dimensionality Reduction is the process of reducing the number of input features (variables) while retaining as much important information as possible.

Goal

To simplify the dataset by reducing unnecessary or redundant features.

Why is it Needed?

Imagine a student dataset with **100 features**, such as:

- Age
- Height
- Weight
- Attendance
- Family Income
- Study Hours
- Sleep Hours
- Internet Usage
- Exam Scores
- ... and many more

Some features may provide similar information.

Instead of using all 100 features, dimensionality reduction may reduce them to **10 important features** without losing much information.

Example

Original dataset:

Height Weight BMI Waist Size Fitness Score

Since Height, Weight, BMI, and Waist Size are related, they can be summarized into fewer features while preserving most of the useful information.

Benefits

- Faster model training
- Less memory usage
- Reduced noise
- Better visualization
- Avoids overfitting
- Improves performance

Applications

- Face recognition
- Image compression
- Medical data analysis
- Text mining
- Gene analysis
- Data visualization

Popular Dimensionality Reduction Algorithms

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)

3. Association Rule Learning

Definition

Association Rule Learning discovers relationships between items that frequently occur together in a dataset.

Goal

To identify "if then" relationships between items.

Example: Supermarket Shopping

Suppose a supermarket has transaction records:

Customer	Items Purchased
1	Bread, Butter, Milk
2	Bread, Butter
3	Milk, Eggs
4	Bread, Butter, Eggs

The algorithm discovers:

If a customer buys Bread, they are also likely to buy Butter.

This is called an **association rule**.

Association Rule Format

Rules are written as:

$$A \rightarrow B$$

Meaning:

If item A is purchased, item B is likely to be purchased as well.

Example:

- Bread $\rightarrow$ Butter

- Mobile Phone → Phone Cover
- Printer → Ink Cartridge

Important Measures

1. Support

Support tells how often an item or rule appears in the dataset.

Example:

If **Bread and Butter** appear together in **40 out of 100 transactions**, then:

$$\text{Support} = \frac{40}{100} = 40\%$$

2. Confidence

Confidence measures how often the rule is true.

Example:

If **50 people bought Bread**, and **40 of them also bought Butter**, then:

$$\text{Confidence} = \frac{40}{50} = 80\%$$

Meaning:

80% of customers who bought Bread also bought Butter.

3. Lift

Lift measures how much stronger the relationship is compared to random chance.

- Lift > 1 → Positive association
- Lift = 1 → No association

- $Lift < 1 \rightarrow$ Negative association

Applications

- Product recommendation systems
- Market basket analysis
- Website recommendation
- Healthcare diagnosis

Summary of Unsupervised Learning Techniques

Technique	Purpose	Example
Clustering	Groups similar data into clusters	Group customers by shopping behavior
Dimensionality Reduction	Reduces the number of features while retaining important information	Compress image data or reduce 100 features to 10
Association Rule Learning	Finds relationships between items that frequently occur together	Customers buying Bread also buy Butter

Comparison of the Three Techniques

Clustering	Dimensionality Reduction	Association Rule Learning
Groups similar data	Reduces the number of features	Finds relationships among items
Output is clusters	Output is fewer features	Output is association rules
Example: Student groups	Example: Compress image data	Example: Bread $\rightarrow$ Butter
Used for segmentation	Used for simplifying data	Used for recommendation systems

Reinforcement Learning (RL)

What is Reinforcement Learning?

Reinforcement Learning (RL) is a type of Machine Learning in which an **agent learns by interacting with an environment**. Instead of learning from labeled examples, the agent **tries different actions, receives rewards or penalties (feedback), and gradually learns the best strategy to maximize the total reward**.

Definition

Reinforcement Learning is a machine learning technique where an agent learns the best actions through trial and error by receiving rewards for good actions and penalties for bad actions.

How Reinforcement Learning Works

Unlike supervised learning (where correct answers are provided), reinforcement learning **does not know the correct answer initially**. The agent learns through **experience**.

Components of Reinforcement Learning

1. Agent

The **agent** is the learner or decision maker.

Examples:

- Robot
- Self driving car
- Game playing AI
- Drone
- Software chatbot

2. Environment

The **environment** is everything with which the agent interacts.

Examples:

- Chess board
- Road for a self driving car
- Robot's surroundings
- Video game

3. State (S)

A **state** describes the current situation of the agent.

Examples:

For a robot:

- Robot is at position (2,3)

For a chess game:

- Current arrangement of pieces

For a self driving car:

- Speed
- Traffic
- Lane position

4. Action (A)

An **action** is a decision taken by the agent.

Examples:

Robot:

- Move Left
- Move Right
- Move Forward

Self driving car:

- Accelerate
- Brake
- Turn Left
- Turn Right

Chess AI:

- Move Queen
- Move Knight

5. Reward (R)

A **reward** is feedback received after taking an action.

- Positive reward → Good action
- Negative reward (penalty) → Bad action

Example:

Robot reaches destination:

Reward = +100

Robot hits wall:

Reward = 50

6. Policy (π)

A **policy** is the strategy the agent follows to choose actions.

It answers:

"What action should I take in this state?"

A good policy maximizes the total reward over time.

Step by Step Example: Robot in a Maze

Imagine a robot trying to reach the exit.

Step 1

State:

Robot starts at the entrance.

Action:

Moves Right.

Reward:

0 (neutral)

Step 2

Moves Forward.

Hits a wall.

Reward:

10

Robot learns:

"Avoid this path."

Step 3

Moves Left.

Moves Forward.

Finds the exit.

Reward:

+100

Robot learns:

"This is a good path."

After many trials, the robot learns the shortest and safest route to the exit.

Example: Self Driving Car

State

- Speed
- Distance to other cars
- Traffic signal

Action

- Accelerate
- Brake
- Turn

Reward

- Safe driving = +10
- Reaches destination = +100
- Accident = -100
- Traffic violation = -50

The car continuously improves its driving policy to maximize safety and efficiency.

playing thousands or millions of games, the AI learns strong strategies.

Exploration vs. Exploitation

One of the key ideas in reinforcement learning is balancing **exploration** and **exploitation**.

Exploration

The agent tries **new actions** to discover whether they lead to better rewards.

Example:

A robot explores a new path in a maze.

Exploitation

The agent uses the **best known action** based on what it has already learned.

Example:

The robot follows the shortest known path to the exit.

A good RL algorithm balances both to continue learning while making good decisions.

Applications of Reinforcement Learning

Application	Example
Robotics	Robot navigation and manipulation
Self driving cars	Steering, braking, lane changes
Game playing	Chess, Go, Atari games
Healthcare	Personalized treatment strategies
Finance	Portfolio optimization and trading
Recommendation systems	Personalized content suggestions
Traffic signal control	Optimizing traffic flow
Industrial automation	Process optimization

Advantages

- Learns through experience.
- Does not require labeled training data.
- Can solve complex sequential decision problems.
- Continuously improves with interaction.
- Suitable for dynamic environments.

Limitations

- Requires many interactions (training can be slow).
- Designing an effective reward function can be difficult.
- Exploration can be computationally expensive.
- Training may require significant computing resources.

Difference Between the Three Types of Machine Learning

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data	Labeled	Unlabeled	No labeled data; learns from interaction
Goal	Predict output	Find hidden patterns	Learn the best sequence of actions
Feedback	Correct answers are provided	No feedback	Rewards and penalties
Output	Class or numerical value	Clusters, reduced dimensions, or association rules	Optimal policy (best action strategy)
Example	Predict house price	Customer segmentation	Robot learning to navigate a maze